



# Demo Script — AI-First CRM HCP Log Interaction

A ready-to-record 10–15 min walkthrough: frontend tour · all 6 LangGraph tools · code explanation · task summary. — Alok Deep

## Before you hit record (2-min checklist)

- ☐ `docker compose up -d` (Postgres running) and backend `uvicorn` up on :8000
- ☐ Frontend `npm run dev` up on :5173
- ☐ **Avoid the rate limit mid-recording:** record just after your Groq daily quota resets, or temporarily set `GROQ_MODEL=llama-3.1-8b-instant` in `backend/.env` (higher free limit).
- ☐ Allow **microphone** permission in the browser once (for the voice demo).
- ☐ Browser zoom so the form (left) **and** chat (right) are both visible.
- ☐ Second terminal ready with: `docker exec -it hcp_crm_db psql -U crm -d hcp_crm`
- ☐ Do one silent dry-run of the prompts below so nothing surprises you on camera.

## The prompts (copy/paste ready — fresh scenario)

| Step | Tool                                    | Prompt                                                                                                                                                                                                                                                                                            |
|------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <code>log_interaction</code>            | Yesterday I had a meeting with Dr. Emily Chen and we went over CardioBoost's efficacy and dosing. She was quite positive. I shared the clinical summary deck and left 2 sample packs. Outcome: she'll trial it with two patients. Follow-up: send the dosing guide and check back in three weeks. |
| 2    | <code>edit_interaction</code>           | Actually that was a video call, not a meeting, and the sentiment was neutral.                                                                                                                                                                                                                     |
| 3    | <code>summarize_notes</code><br>(voice) | <i>spoken — see script below</i>                                                                                                                                                                                                                                                                  |
| 4    | <code>suggest_followups</code>          | What follow-up actions do you recommend for this visit?                                                                                                                                                                                                                                           |
| 5    | <code>search_hcp</code>                 | Pull up Dr. Sharma from the CRM.                                                                                                                                                                                                                                                                  |
| 6    | <code>save_interaction</code>           | Great, save this interaction to the database.                                                                                                                                                                                                                                                     |

### Voice-note line to speak (step 3):

"So I just wrapped up with Dr. Chen — we spent most of the time on the safety profile. She had concerns about drug interactions in elderly patients, so I walked her through the contraindications. She seemed reassured but wants more data before committing."

## Timestamped script

### 0:00 – 1:00 · Intro + what the task is

"Hi, I'm Alok. This is my AI-First CRM — the **Log HCP Interaction** screen for pharmaceutical field reps. The idea: a rep shouldn't waste selling time typing forms. On the **left** is the structured interaction form; on the **right** is an AI assistant. The key rule from the brief is that I **don't fill the form manually** — I describe the visit in natural language and a **LangGraph agent** powered by **Groq** extracts everything and fills the form. It has **six tools**, and I'll demo all of them."

### 1:00 – 2:15 · Frontend walkthrough

Point out the split-screen, the **Google Inter** font, the form fields (HCP name, type, date/time, attendees, topics, materials, samples, sentiment, outcomes, follow-ups), and the chat panel.

"Everything's built in **React with Redux Toolkit** — the form and chat share Redux state. When the AI updates a field, you'll see it **flash-highlight**, and the assistant shows a **badge** naming the exact tool it called. Let me show you."

## 2:15 – 3:45 · Tool 1 — log\_interaction

Paste **Prompt 1**. Let it fill.

"One sentence, and watch: it extracted the **HCP name**, resolved '**yesterday**' to a real date, set the **type**, wrote a professional **Topics Discussed** summary, picked up **positive sentiment**, the **materials**, the **samples**, the **outcome**, and the **follow-up action** — each highlighted. The badge confirms **log\_interaction** fired. This tool is doing LLM **entity extraction and summarization**."

## 3:45 – 4:45 · Tool 2 — edit\_interaction

Paste **Prompt 2**.

"Now I made a mistake. Instead of clicking the form, I just tell it. Notice it changed **only** the interaction type to Virtual and the sentiment to Neutral — everything else stayed exactly as it was. That's **edit\_interaction** : targeted field updates, not a re-log."

## 4:45 – 6:15 · Tool 3 — summarize\_notes (voice note)

Click 🔊 **Summarize from Voice Note**. Speak the voice line. Click ■ **Stop**.

"This is the voice workflow — and notice the label says **Requires Consent**, because recording an HCP conversation legally requires it. I record, and the audio goes to **Groq's Whisper** model for transcription, then the **summarize\_notes** tool condenses it into clean Topics Discussed bullets. So that's a **third Groq model** working with the chat LLM."

Show the transcribed + summarized result.

## 6:15 – 7:00 · Tool 4 — suggest\_followups

Paste **Prompt 4**.

"I can ask for next steps. **suggest\_followups** reads the whole interaction and proposes three concrete actions — you can see them under **AI Suggested Follow-ups**."

## 7:00 – 7:45 · Tool 5 — search\_hcp

Paste **Prompt 5**.

"This one hits the **database**. **search\_hcp** looks Dr. Sharma up in Postgres and, since there's a single match, sets it on the form. In a real CRM this pulls from your HCP master list."

## 7:45 – 8:45 · Tool 6 — save\_interaction + prove it in Postgres

Paste **Prompt 6**. Then switch to the psql terminal:

```
SELECT id, hcp_name, interaction_type, sentiment FROM interactions;
```

"And **save\_interaction** writes the record to **PostgreSQL**. Here's the row in the actual database — so the full loop, from spoken/typed input to a persisted CRM record, is real."

## 8:45 – 11:30 · Code walkthrough

Open these files and narrate briefly:

- `backend/app/agent/graph.py` — "This is the LangGraph **StateGraph**. It's an **agent → tools → agent** loop: the agent node is the Groq LLM with tools bound; if it calls tools, the **ToolNode** runs them and control comes back for a final reply."
- `backend/app/agent/tools.py` — "The six tools. Each is a typed **@tool**; form-mutating ones return a **Command** that patches shared state. **log\_interaction** and **edit\_interaction** are mandatory; **summarize\_notes** and **suggest\_followups** call the LLM themselves; **search\_hcp** and **save\_interaction** touch Postgres."
- `backend/app/agent/state.py` — "Shared state with reducers that merge each tool's field-level deltas."
- `backend/app/routers/chat.py` — "Two endpoints: **/api/chat** runs the agent, **/api/transcribe** runs Whisper."
- `frontend/src/store/` — "Redux slices: **interactionSlice** holds the form and applies AI updates; **chatSlice** has the **sendMessage** thunk that posts to the backend and dispatches the returned **form\_updates**."
- `frontend/src/components/` — "**InteractionForm** and **ChatPanel**, plus the **useRecorder** hook for the mic."

"So the frontend owns the form, sends its snapshot each turn, and the backend returns only what changed."

### 11:30 – 12:45 · Architecture + stack + the model note

"Stack: **React + Redux, FastAPI, LangGraph, Groq** (Llama 3.3 70B + Whisper), **PostgreSQL, Inter** font. One honest note on the LLM: the brief named `gemma2-9b-it`, but **Groq decommissioned that model in October 2025**. The same brief also permits `llama-3.3-70b-versatile`, which is active and much better at multi-tool calling — and it's a one-line env change, so any Groq model drops in."

### 12:45 – 14:00 · Task summary (what I understood)

"To summarize what I took from the task: the goal was an **AI-first** CRM interaction screen where the agent — not manual data entry — drives the form, using **LangGraph** and an **LLM** as mandatory, with at least five tools including Log and Edit. I approached it as a life-science tool: consent-aware voice capture, sentiment, samples, and follow-ups that map to how a real field rep works. The result is six working tools, a persistent database, and a conversational UI that fills and corrects itself. Thanks for watching."

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## If something misbehaves on camera

- **429 rate limit** → you hit the Groq daily cap; switch to `llama-3.1-8b-instant` and restart, or wait for reset.
- **A field doesn't fill** → just restate it more explicitly; the agent will `edit_interaction` it.
- **Mic won't record** → use the  **Paste** button next to the record button and paste the voice line instead.